

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50
No. of Questions: 1

Time: 60 Mins

Annexure A Agent Design Assessment

Code of Conduct:

I Kamalli shree V(192525030) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

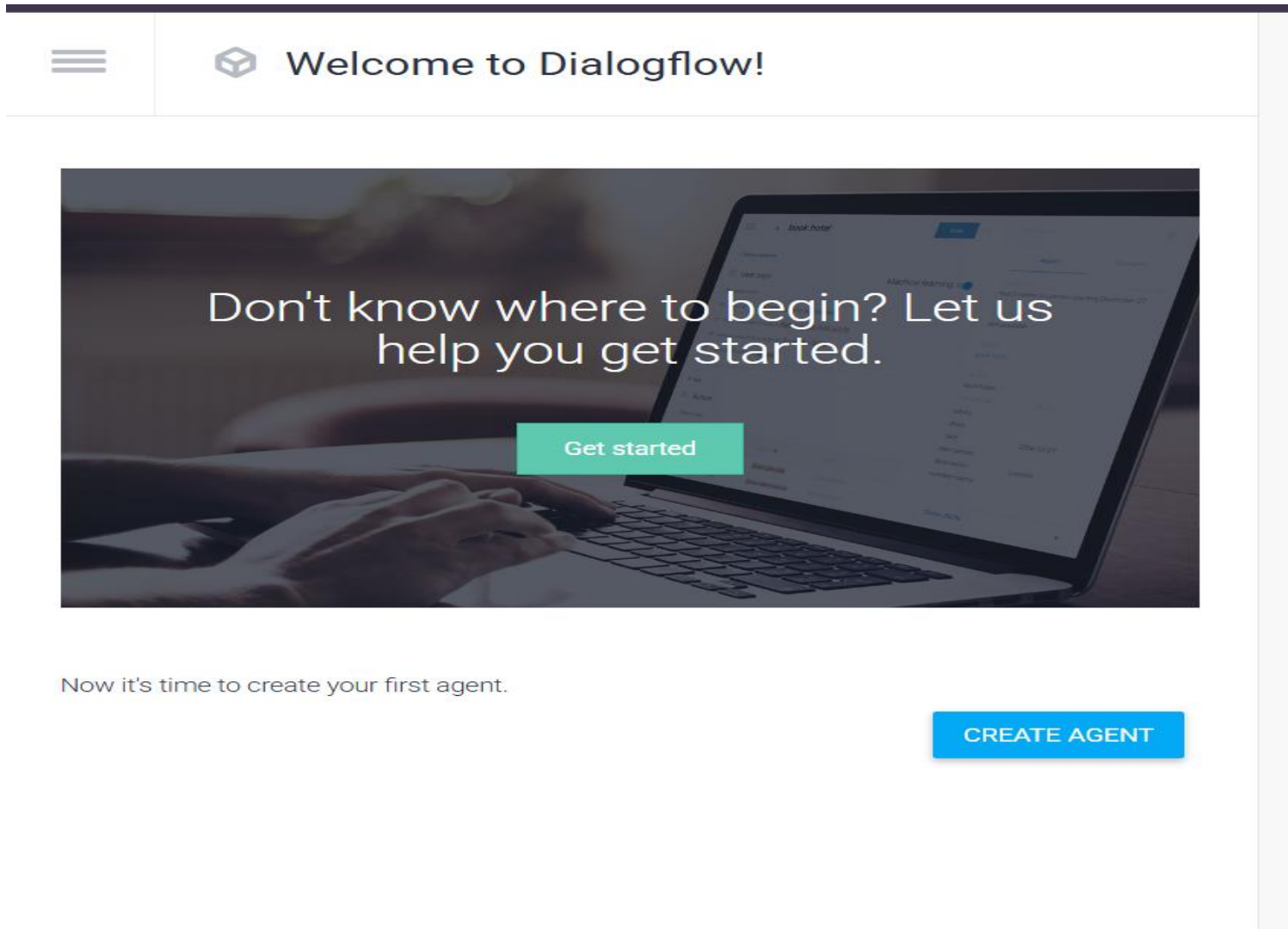
CO1: ASSESSMENT TOOL 1

Name : Kamalli shree V

Reg. No.: 192525030

Department: AI & ML

Course: MLA 0103- Artificial intelligence and expert systems





StudentSupportAssistant

CREATE

DEFAULT LANGUAGE

English — en



Primary language for your agent. Other languages can be added later.

DEFAULT TIME ZONE

(GMT+6:00) Asia/Almaty



Date and time requests are resolved using this timezone if not provided in the API requests.

GOOGLE PROJECT

Create a new Google project



Enables Cloud functions, Actions on Google and permissions management.

AGENT TYPE



Set as Mega Agent

Combine multiple Dialogflow agents (i.e. sub agents) into a single agent (i.e. [mega agent](#)).



Intents

[CREATE INTENT](#)

Default Fallback Intent



Default Welcome Intent

Add follow-up intent



No regular intents yet. [Create the first one.](#)

Intents are mappings between a user's queries and actions fulfilled by your software. [Read more here.](#)

Before you start, check out [Prebuilt Agents](#), a collection of agents developed by the Dialogflow team.



• Course Registration

SAVE



Contexts ?



Events ?



Training phrases ?



Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.



Train the intent with what your users will say

Provide examples of how users will express their intent in natural language. Adding numerous phrases with different variations and parameters will improve the accuracy of intent matching. [Learn more](#)

[ADD TRAINING PHRASES](#)

Training phrases

Search training phrases 



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” I want to enroll

” Open registration



” Registration process

” Register for semester

” Course registration

” How can I register?

Responses



DEFAULT +

Text Response



- | | |
|---|---|
| 1 | Course registration opens at the beginning of every semester. Please login to the Student Portal and complete the registration before the deadline. |
| 2 | Enter a text response variant |

ADD RESPONSES

Intent saved

OK



Exam Schedule

SAVE



Training phrases

Search training phrases  



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” Add user expression

” Upcoming exams

” Exam timetable

” Semester exams

” When is my exam?

” Exam dates

Responses



DEFAULT 

Text Response



- 1 The examination schedule is available on the university website under the Examination section.
- 2 Enter a text response variant

ADD RESPONSES



Attendance

SAVE



Contexts ?



Events ?



Training phrases ?

Search training phrases



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” Add user expression

” check attendance

” attendance details

” My attendance

” Attendance percentage

Text Response



1 You can view your attendance percentage through the Student Portal.

2 Enter a text response variant

ADD RESPONSES



Set this intent as end of conversation ?



• Timetable

SAVE



Training phrases

Search training phrases  



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When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

” Add user expression

” show classes

” time table

” class schedule

” **Today's** timetable

Responses



DEFAULT +

Text Response



1 Your timetable can be accessed through the Academic Portal.

2 Enter a text response variant

ADD RESPONSES

☐ Set this intent as end of conversation 



Campus Services

SAVE



Training phrases

Search training phrases  



Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

” Add user expression

” wifi support

” medical centre

” transport

” hotel office

” library things

Responses



DEFAULT +

Text Response



1 Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office|

2 Enter a text response variant

ADD RESPONSES



@services

SAVE



Define synonyms ?



Regex entity ?



Allow automated expansion



Fuzzy matching ?



Separate synonyms by pressing the **enter**, **tab** or **;** key.



Library	Library	
Hostel	Hostel	
Transport	Transport	
Medical	Medical	
WiFi	WiFi	Enter synonym

Agent

USER SAYS

[COPY CURL](#)

hi



DEFAULT RESPONSE



Greetings! How can I assist?

INTENT

[Default Welcome Intent](#)

ACTION

input.welcome

Agent

USER SAYS

[COPY CURL](#)

library timings



DEFAULT RESPONSE



Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office.

INTENT

[Campus Services](#)

ACTION

Not available

USER SAYS

[COPY CURL](#)

how do i register?



DEFAULT RESPONSE



Course registration opens at the beginning of every semester. Please login to the Student Portal and complete the registration before the deadline.

INTENT

[Course Registration](#)

Agent

USER SAYS

[COPY CURL](#)

attendance



DEFAULT RESPONSE



You can view your attendance percentage through the Student Portal.

INTENT

[Attendance](#)

ACTION

Not available

USER SAYS

[COPY CURL](#)

Who won the IPL?



DEFAULT RESPONSE



Say that one more time?

CONTEXTS

[RESET CONTEXTS](#)

INTENT

[Default Fallback Intent](#)

ACTION

input.unknown

USER SAYS

[COPY CURL](#)

show timetable



DEFAULT RESPONSE



Your timetable can be accessed through the Academic Portal.

INTENT

[Timetable](#)

ACTION

Not available

PARAMETER

VALUE

date-time

USER SAYS

[COPY CURL](#)

exam schedule



DEFAULT RESPONSE



The examination schedule is available on the university website under the Examination section.

INTENT

[Exam Schedule](#)

ACTION

Not available



• Default Welcome Intent

SAVE



[+ New parameter](#)

Responses ?



DEFAULT +

Text Response



- | | |
|---|--|
| 1 | Hi! How are you doing? |
| 2 | Hello! How can I help you? |
| 3 | Good day! What can I do for you today? |
| 4 | Greetings! How can I assist? |
| 5 | Welcome to the Student Support Assistant.
How may I help you today? |
| 6 | Enter a text response variant |

ADD RESPONSES

USER SAYS

[COPY CURL](#)

course registration



DEFAULT RESPONSE

Course registration opens at the beginning of every semester. Please login to the Student Portal and complete the registration before the deadline.

INTENT

[Course Registration](#)

USER SAYS

[COPY CURL](#)

exam timetable



DEFAULT RESPONSE

The examination schedule is available on the university website under the Examination section.

INTENT

[Exam Schedule](#)

USER SAYS

[COPY CURL](#)

attendance



DEFAULT RESPONSE

You can view your attendance percentage through the Student Portal.

INTENT

[Attendance](#)

USER SAYS

[COPY CURL](#)

library timings



DEFAULT RESPONSE

Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office.

INTENT

[Campus Services](#)

USER SAYS

[COPY CURL](#)

transport



DEFAULT RESPONSE

Campus services include Library, Hostel, Transport, Medical Centre and Student Support Office.

INTENT

[Campus Services](#)